Body Language Communication

Derek Chadee • Aleksandra Kostić
Editors

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To Dr. Hamid Farabi, for his spiritual guidance, friendship and love.
And to my darling, loving and ever-caring wife, Mary Chadee (DC)
Paul Ekman, a great scientist and friend who awakened
my interest in the field of nonverbal behavior (AK)

Beyond Speech: The Dynamics of Body Language

In 1970, just over a half-century ago, Julius Fast published a well-received commercialized book titled Body Language (Fast 1970). Though this book popularized body language and created international curiosity about this emerging topic, the science, literature, and precision in understanding body language have developed many folds since then. Today, tremendous literature and material on this engaging area elucidate more than mystify as fifty years ago.

People are social beings who communicate with others emotionally, expressing themselves not only verbally but powerfully with their bodies via facial expressions, gestures, and postures. They choose the verbal or non-verbal channel to articulate their cognition and affect states voluntarily or involuntarily. Though people are interested in the quality of their communication, they often do not understand the functioning of their communication channels, especially the nonverbal ones.

Nonverbal signals are elements of behaviour emitted by one person, which then affect the behaviour of the person receiving them (Argyle, 1988). Unlike pre-planned, precisely thought out, and organized speech, nonverbal messages are perceived as more spontaneous, sincere, and reliable because they are less controlled than verbal ones and more trusted (Rot, 2004). The often vague link between nonverbal signals and their meanings can shield sensitive or intimate feelings that a person unconsciously conveys but may not be ready to express clearly and directly in words. This is especially true when it comes to communicating emotions. Nonverbal signals, such as facial expressions, touch, eye contact, gestures, body posture, and tone of voice, convey a wide range of messages about emotions, interpersonal attitudes, and a person's sincerity. These signals add complexity to interactions, requiring strong communication skills from people. For effective communication, individuals must stay focused and attuned to the nuances of the interaction, paying attention to the body language and nonverbal cues and their meanings at the appropriate moments.

During a given communication, parties strive to convey a relatively general impression of their interlocutor, relying not only on their words but also on facial expressions, tone of voice, and body movements. Communication with other people should create an atmosphere of mutual trust, benevolence, an open and honest exchange of information, the experience of new perspectives and positive expectations. Nonverbal signals and their messages are important in realizing goals within interaction. Of course, they must be adequately used and correctly interpreted within the framework of a given relationship and a social context.

Understanding body language provides insights into a dimension of social interaction that, even after half a century, there are more questions than answers. As a supplement to speech, body language often talks louder than words. As conveyors, our nonverbal behaviour expresses many emotions, even if the social situation may suppress their expression. Though body language is present in all social situations, understanding it is crucial during negotiation, persuasion, expression of leadership, conflict resolution, and presentation of ourselves. The paradigm of body language and nonverbal behaviour has grown over the last five decades. As this paradigm evolves, new theories, research, and questions emerge that require serious consideration. This volume addresses some of the emerging concerns and issues in the body language and nonverbal behaviour paradigm.

The volume covers a wide range of areas within the body language literature, intending to add further insights, address debates, and meaningfully contribute to the existing literature. Among the topics covered in the book are hand gestures and communication, touch and emotional connections, contextual influences on facial expressions, voice and nonverbal expression, posture in social relationships, critiques of body language analyses, personality and nonverbal behaviour, expressive body movements, emotions in movements, postures, and dance, non-facial emotional expressions and conflict resolution, facial thermography and emotions, physical manifestations of thoughts and feelings, nonverbal behaviour in investigative interviewing, lie detection, complexities of facial expressions, relational dominance and predictability, and human-machine nonverbal interaction.

Kinesics, often thought of as body language, is a communication code with a wide range of nonverbal communications through gestures and movements. The chapter by Judee K. Burgoon, Norah E. Dunbar, Steven Pentland, Xinran Wang, Xunyu Chen, and Saiying Ge explores how non-verbal cues communicate credibility, trust, and deception. While facial expressions are often highlighted in nonverbal communication studies, the chapter expands the focus to include channels like vocalics, kinesics, and proxemics. They note that the body is a rich source of communication about everything from one's identity, social status, and emotional state, to self-presentation, credibility, social influence, and veracity. The authors explore the importance of facial expressions and discuss other nonverbal channels like voice, gestures, and eye movements. After summarizing some of the more prominent literature about nonverbal communication from these kinesic and vocalic channels, three studies illustrate how these cues can assess credibility, composure, and trustworthiness in various situations. The chapter concludes by discussing applications in areas like interview training and business negotiations.

Our hands, often taken for granted, play a surprisingly complex role in how we communicate, learn, and experience the world around us. John White and David Givens' chapter probes into the fascinating role of the hands, exploring how hands have evolved to enable us to grasp objects and shape our thoughts and interactions. The journey begins with the neurological underpinnings of nonverbal communication through hand gestures, a skill rooted in ancient neural pathways. The authors note that human use of their hands to send nonverbal messages is rooted in ancient neurocircuits of the posterior hindbrain and anterior spinal cord. The chapter traces the development of the hand from an infant's early reflexes to the refined precision grip by the age of one. The chapter also explores the connection between hand movements and cognition. The concept of embodied cognition, explaining how gestures can influence memory and communication, is discussed. Touch is one of our earliest sensory experiences that plays a crucial role in both childhood development and adult relationships. The various messages conveyed through common hand movements, from the ubiquitous handshake to hands-on-hip posture, are explored. By weaving together evolutionary biology, developmental psychology, and cultural influences, the chapter paints a rich portrait of our hands, not just as tools but as fundamental aspects of what makes us human. The authors concluded that hands are integral to interacting with and learning about the world around us. With deep evolutionary roots, hands are hard-wired in how humans engage with the world.

Jonathan Bowman and Benjamin Compton's chapter stimulatingly investigates touch in nonverbal emotional communication, highlighting how haptic interactions can influence relationships. They note that touch as a form of nonverbal expression is known as *haptic messaging* (often shortened to "haptics" as a stand-in for the longer expression). While much of the existing research emphasizes facial expressions, this chapter argues that touch conveys emotional messages with unique potency. From the initial interactions between strangers that can set the tone for their relationship to the nuanced expressions of affection, openness, or even anger in established connections, touch emerges as a powerful nonverbal code. The chapter explores various structural forms and functional expressions of touch, examining how they convey emotions and affect both the giver and receiver. Additionally, Expectancy Violation Theory is applied to understand how touch impacts relationships in different contexts, offering insights into the dynamics of both anticipated and unexpected tactile interactions.

Context plays a crucial role in how we perceive and interpret our reality, including facial expressions. Sam Day, Danielle Shore and Eva Krumhuber's chapter focuses on the importance of interpreting facial expressions within social interactions, highlighting that accurately decoding these expressions is complex due to the influence of various contextual cues, such as body language. They posit that decoding a social partner's emotions is fundamental to initiating and maintaining social interactions and interpersonal relationships. The chapter reviews empirical research on how these contextual factors affect the interpretation of facial expressions and vice versa. They advocate for a holistic and contextualized approach in future research in affective and social psychology to better understand the interplay between facial expressions and their surrounding context in social and emotional judgments.

Terrence Horgan, Judith Hall, and Nathaniel Miller's chapter on nonverbal expression in voice acknowledges that the human voice expresses emotions through variations in sound elements like pitch, volume, timbre, tone, rhythm, and speed, which indicate how people feel and their physical and mental conditions. The chapter moves forward to discuss the implications of the human voice in disorders such as alexithymia, which affects emotional awareness, and Parkinson's disease, which impacts vocal function, can hinder this form of communication. Conversely, psychiatric conditions like depression and mania might amplify emotional expression through the voice. The chapter also provides insights for caregivers to understand better and respond to the emotional needs of those with these conditions.

Hallee Altman and Stephen Nowicki provide a very intriguing understanding of posture and the formation of social relationships. Their chapter explores the role of posture as a key aspect of nonverbal communication within the Nowicki and Duke four-phase model of relationship and its impact on relationships throughout various stages, including initiation, development, and conclusion. The authors argue that posture is often undervalued in discussions about relationship dynamics and provide evidence supporting its significance. They advocate for educational programmes to enhance individuals' postural communication skills and outline a potential approach for such training. The chapter concludes by emphasizing the need for further research to better understand and improve postural communication abilities in both children and adults. Despite extensive research on other nonverbal cues, posture remains underexplored but is vital in conveying authentic emotions.

Vincent Denault, Miles Patterson, Mircea Zloteanu, and Victoria Talwar's chapter critically assesses body language and the associated analyses. The term "body language" has been commercialized since the 1970s, with the emergence of numerous training programmes and a plethora of materials, including books and videos on understanding and interpreting body movements. The rise of social media has further fuelled this industry by spreading body language content to an unprecedentedly broad audience.

However, Denault and colleagues' aim is to demystify the concept of body language, asserting that no consistent or clear-cut interpretations can be reliably drawn from specific nonverbal behaviours. They address common misconceptions associated with body language analysis, critically evaluating the effectiveness of training programmes designed to teach the "codes" for deciphering facial and bodily expressions. They note that a central theme since the 1970 in body language literature is that there are codes to understand what others say with their bodies. Finally, the belief that "the body never lies," a core tenet in many body language training courses and publications, is debunked. The chapter concludes by offering guidance on critically assessing the claims made by body language experts. The chapter is a refreshing critical commentary on body language.

What is the relationship between nonverbal bodily behaviour and personality in social interaction? This is the core question that Nora Murphy addresses in her exceedingly insightful chapter. At the start of her chapter, Murphy posits that nonverbal behaviour occurs automatically and is vital to social perception processes and personality traits, profoundly impacting individual and social outcomes. The chapter addresses the following crucial areas (1) the role of nonverbal behaviour and personality in social perception processes; (2) how observable nonverbal behaviour is fundamental to defining personality; (3) documented nonverbal bodily behaviours associated with specific personality traits; (4) whether descriptions of personality include bodily behaviours; and (5) whether nonverbal bodily behaviours are captured in personality inventories. She concludes with a simple but profound statement: "Personality and nonverbal *bodily* behaviour are not linked in simple ways during social interactions."

R. Thomas Boone and Joseph Cunningham's chapter explores the research on the emotional meaning conveyed through expressive body movement, covering various forms such as dance, gait, posture, and dynamic movement. Despite the limited research in this area, their chapter supports the view that body movement is a robust nonverbal channel for communicating emotions. It matches the effectiveness and developmental alignment of other nonverbal channels like facial expressions, vocal intonations, and music. The chapter aims to provide an overview of existing research, discuss the challenges specific to studying body movement, highlight overlooked theoretical connections, and suggest directions for future research to deepen our understanding of how body movements convey emotions. In concluding, they state, "Recent efforts have identified a methodology for coding these behaviours, it is now time to refine that methodology via a more theoretically driven approach to establishing emotion-specifying cue structures and perceptions."

Alan Crawley and Ronald Riggio make a useful contribution by noting the evolution and complexity of nonverbal communication and focusing on how body movements and postures express emotions through a probabilistic relationship influenced by natural selection. They argue that emotions are conveyed through various nonverbal cues, each with its efficiency and redundancy in communication. The discussion emphasizes that while certain body movements are generally associated with specific emotions, their interpretation is context-dependent and culturally influenced. The chapter calls for future research to use more realistic methods to understand better how emotions are conveyed through body expressions accurately and reliably. The main arguments of the chapter are: (1) the nonverbal modalities used to express emotions depend on the type of affect felt at the moment, and the efficiency of that modality; (2) the modalities of nonverbal displays of emotions are interrelated such that redundancy is built into the communication system; and (3) evolution has endowed humans with a probabilistic repertoire of behaviours associated with emotions. In concluding, they recognize that body movements and positions are complex and not as straightforward as detecting a posture and instantaneously determining the corresponding emotion.

Our human emotions allow us to engage, coordinate, and manage a range of social interactions and to respond to the daily range of interpersonal conflicts that emerge and must be negotiated. Martin Remland and Tricia Jones, in their chapter on emotional expressions beyond the face, noted that much attention has been paid to the relationship between particular emotions and discrete patterns of facial cues that convey these emotions. However, less attention has been given to the similarity between emotions and vocal and bodily expressions of these emotions. The authors acknowledge that emotional communication is a multi-modal process involving the voice, the whole body, and personal space, gaze and touch. They assess the nature and range of emotions and the role of contexts in the expression of body and vocal communication of human emotions. An observation from the chapter is the importance of vocal cues reflecting emotional states often identified by listeners when there are audible modifications of a speaker's respiration, phonation, and articulation. Identification of emotional states via voice is crucial in interpersonal relationships. The authors' findings applied to how the non-facial expressions of self-conscious emotions influence conflict management interventions.

Dennis Küster's chapter explores what facial thermography can tell us about emotions. While progress in recording and analysing facial image data has been notable, this development has overlooked using facial thermal imaging (FTI) to study facial expressions. The author accentuates facial thermal imaging to reveal underlying emotional states and focuses on the physiological basis and ability to distinguish between different facial regions of interest (ROIs) using FTI. In addition, the author emphasizes the opportunities and challenges in integrating FTI with other modalities and biosignals. The author recommends that such integration provides a more robust approach to data for modelling real-time adaptive systems and emotions. However, it is recommended to develop ethical adaptive systems that leverage thermal imaging data to benefit humans while protecting their privacy and autonomy. This chapter provides an interesting and what may appear to be a rare angle to understanding nonverbal behaviour and beckons body language research into a dimension that is sparse in the nonverbal communication literature.

Mark Frank, Zachary Glowacki, Madison Neurohr, Ifeoma Nwogu, and Anne Solbu's chapter highlights how the human body communicates messages from its appearance and movements via the processes of our thoughts and feelings. As they observe, our bodies' dynamic movement, fast, voluntary and involuntary, sends messages that reflect our thoughts and feelings. Their chapter examines Darwin's theory of evolution, emphasizing the principle of serviceable habits, antithesis, and nerve force on the body's physiological responses. The authors suggest we communicate messages through postural actions, forward, backwards, up and out or down and in, as well as hand and head movements. Therefore, our emotions and cognitive processes manifest in body movements such as touch, gait, and gestures. The discussion emphasizes the convergences of complex interactive situations and emotional processes that influence body behaviour. As they note, body movements are not just social signals but can also be instrumental in survival in both a distal and a proximal sense. The authors conclude the chapter by discussing the challenges to coding bodies using machine-based approaches.

Mary Chadee, Derek Chadee, and Aleksandra Kostic assess how investigative interviews may be affected by the interpretation of nonverbal behaviour and body language. Investigative interviews aim to elicit accurate information, yet nonverbal communication poses a complex challenge. The authors observed that investigators are often trained to look for nonverbal cues at the start of an interview to ascertain truthful nonverbal behaviour, which is then used as a baseline to detect deceit. However, the perception and interpretation of body language in these interviews are influenced by various factors. These include psychological reactance, cognitive schemas, interviewer bias, the fundamental attribution error and how the brain's amygdala and prefrontal cortex respond to high anxiety situations such as those that exist in an investigative interview as they are intrusive. In conclusion, the authors note that future research and theorization must consider an investigative interview's social and psychological dynamics. There is a need to understand the power relationship between the interviewer and the interviewee and the dominance and subordination messages that may be present as the interview progresses.

Aldert Vrij and Ronald Fisher's chapter examines detecting deception from nonverbal signals, stating that the interviewing process is cognitively demanding and physical signals can point towards deception or evidence of truth. However, the authors draw attention to seminal research in deception that highlights the non-universality of nonverbal lying cues, emphasizing deception detection techniques like microexpressions of emotions and the behaviour analysis interview (BAI). They further discuss several reasons why nonverbal cues to deception are unreliable, stating that both lie and truth-tellers employ the same nonverbal strategy. They will show similar signs of increased nervousness due to the fear of not being believed. The chapter concludes by discussing the future of nonverbal lie detection, focusing on automated lie detection, the development of interview techniques to elicit or enhance nonverbal cues to deception and situations in which analyses of only nonverbal behaviour can be made.

José-Miguel Fernández-Dols and Roberto García-Gonzalez's work on facial expression at the crossroads is a useful contribution to the body language literature. The assumption that there are several universal expressions of evolutionary acquired emotions has undergone methodological and theoretical criticism supported by empirical findings. Hence, with this premise, the authors argue that this area is at a crossroads. Recent advancements call for rethinking the concept of facial expressions. The traditional notion, which views facial expressions as fixed, innate, and independent of context, is being challenged. Modern research emphasizes the adaptability and context-dependency of facial behaviour, suggesting that expressions are not static but influenced by surrounding circumstances and cultural factors. Studies show that situational context can significantly alter the perception of facial expressions, indicating that expressions are not isolated behaviours but part of a broader communicative framework. This leads to the proposal that understanding facial expressions should involve considering their dynamic and interactive nature across different cultures and contexts. Fernández-Dols and García-Gonzalez conclude that the concept of facial expression requires an urgent re-conceptualization that has been simultaneously demanded by three parallel processes—new theoretical hypotheses; new technologies; and the recognition of the growing sensitivity to cross-cultural differences.

"Dominance and submission exist in all relationships" is the starting point of James Honeycutt, Ryan Rasner, Laura Labato, and Dongdong Yang's contribution. The majority of research describes dominance as a trait. However, the authors posit the usefulness of conceptualizing dominance from a state perspective. Both situational dominance and the evolving stages in a relationship influence the body language display and its intensity associated with dominance. Dominance varies depending on how much power an individual wields in a situational context. Some persons are inflexible and context-insensitive in the expression of dominance and submissive patterns. Dominance is reflected through nonverbal cues such as facial expressions, gestures, postures, movement, access to greater personal space, nonreciprocal touch, and eye gaze and paralanguage cues including pitch, volume, pauses, and interruptions. The authors discuss the stages of relational escalation: initiating, experimenting, intensifying, integrating, and bonding. They argue that in viewing dominance as state rather than trait, dominance can be understood as a phenomenon represented by the asymmetry of predictability. During an argument between parties the asymmetry of predictability can be measured through various nonverbal communication cues. The chapter discusses the nonverbal aspects of conflict resolution, mutual vs. unilateral influence, and dominance in terms of nonverbal cues of predictability (e.g., eye gaze, silence, voice loudness). Essentially, the dominant person in an argument has less predictable behaviours, while the followers are more predictable. Honeycutt and colleagues have provided an understanding of dominance and body language that can be extrapolated to various social contexts.

Over the past two decades, there has been a significant expansion in the scope of human-machine interaction (HMI), moving beyond traditional textual and verbal communication to include nonverbal elements. The contribution of Isaac Wang, Jaime Ruiz, and Arvid Kappas is intriguing as they explore the relationship between HMI and nonverbal behaviour. Initial research predominantly centred on facial expressions, but recent advancements have broadened the focus to include "body language," such as gestures, posture, and proxemics. These new developments are reflected in areas such as affective computing and social robotics, which aim to enhance HMI by evaluating and interpreting human behaviours in affective, motivational, and socially pragmatic contexts. Nonverbal cues like gestures, gaze, and facial expressions enrich verbal communication by adding depth, emotion, and context. As technology advances, virtual and embodied agents are increasingly capable of mimicking human nonverbal behaviours. This capability allows machines to interpret user movements and emotions and convey their responses more naturally and effectively. The integration of body language in virtual agents and robots significantly enhances human-machine interactions by facilitating communication, conveying emotions, and improving collaboration. Studies show that nonverbal cues like nodding, mutual gaze, and gestures help agents appear more attentive, human-like, and engaging while supporting efficient task completion and rapport building.

The exploration of body language throughout this book reveals its profound influence on human communication and interaction. From its origins as a commercialized concept to its current status as a complex field, the study of body language has provided an in-depth understanding of conveying emotions, intentions, and social dynamics in social interactions. The chapters address various facets, from facial expressions and gestures to posture and touch, highlighting their roles in shaping perceptions and fostering understanding in interpersonal relationships. Integrating body language into human-machine interactions opens new avenues for more empathetic and effective communication as technology advances. Despite its evolution over the decades, body language remains a dynamic area of study, continually evolving with new research and insights that enrich our understanding of nonverbal communication. As we look ahead, the ongoing exploration of body language promises to uncover further nuances and applications that enhance our ability to connect and communicate in an increasingly complex world.

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Notes on Contributors

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